Generative AI is primed to make an increasingly strong impact on enterprises over the next five years.

- By 2025, 30% of enterprises will have implemented an AI-augmented development and testing strategy, up from 5% in 2021. – *Gartner*
- By 2026, generative design AI will automate 60% of the design effort for new websites and mobile apps. – *Gartner.*
- By 2027, nearly 15% of new applications will be automatically generated by AI without a human in the loop. This is not happening at all today. – *Gartner.*
- Generative AI coding support can help software engineers develop code 35 to 45 per cent faster, refactor code 20 to 30 per cent faster, and perform code documentation 45 to 50 per cent faster. - *McKinsey*
- Forty-four per cent of the workers' core skills are expected to change in the next five years. Training employees to be able to leverage generative AI is going to be critical. – *World Economic Forum*

Key characteristics of generative AI technology

- ***Text management**:* Generative AI can complete a given text in a coherent manner. It can translate text from one language to another, summarise it into a shorter and concise form, and generate text that mimics human writing.
- ***Contextual understanding**:* Generative AI has a strong ability to understand the context.
- ***Natural language processing (NLP)**:* It can perform various NLP tasks — it can understand and process human language, allowing users to ask questions in a conversational manner.
- ***Question answering**:* Generative AI can answer questions based on the knowledge base.
- ***Personalisation**:* Generative AI can be fine-tuned for specific use cases.
- ***Multi-language support**:* Generative AI can perform NLP tasks in multiple languages.
- ***Advanced semantic search***: It enables semantic search through large volumes of structured and unstructured data, including databases, documents, etc. It also supports auto data indexing using custom AI/ML based data processing.
- ***Content moderation**:* It has smart content filtering and a moderation engine for query and response, which is trainable on enterprise data.
- ***Integration:*** Generative AI can be integrated with various applications/data sources within an organisation, including CRM, ERP, and other proprietary systems, to access and analyse data via API.
- ***Role-based access control**:* The system can be set up with role-based access control, ensuring that users can access and query only data that they are authorised to view.
- ***Software coding**:* It helps with code generation, translation, explanation and verification.

Generative AI architectures and datasets

Figure 1 shows the contextual architecture of generative AI with the various stages involved — acquire, process, store, train and deliver.

AI data sources: The data sources provide the insight required to solve

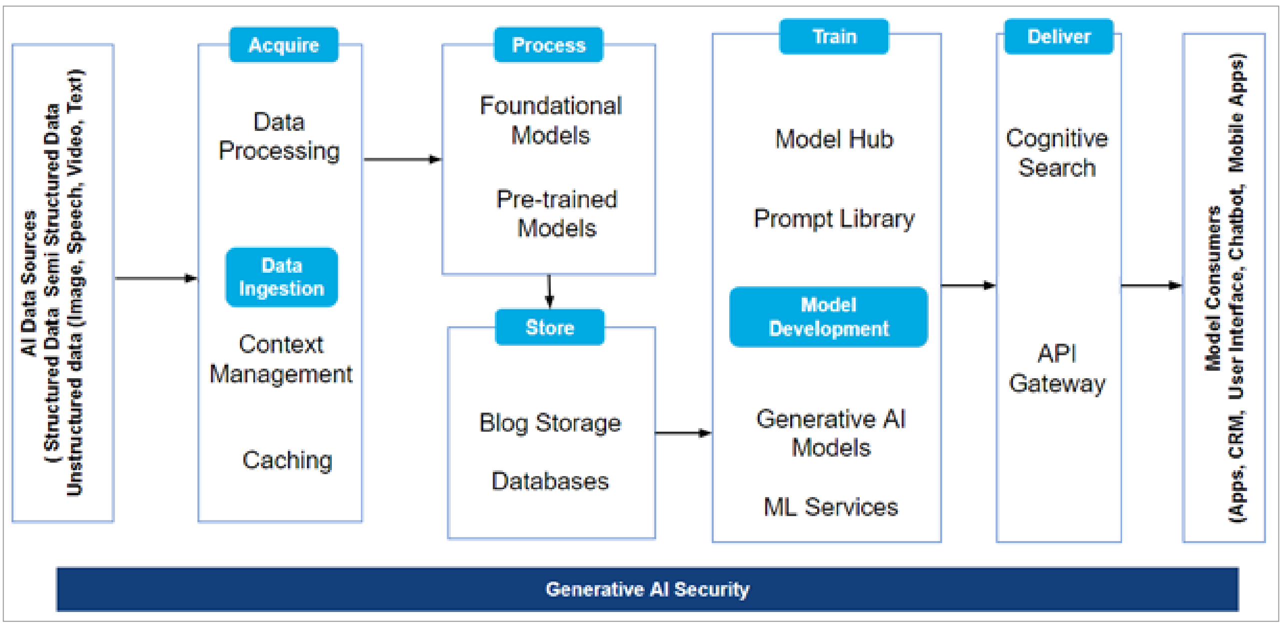

Figure 1: Generative AI architecture